Question 1

Evaluate ∑(x2+2)∑(x2+2) for the sample data {1,4,6,8}{1,4,6,8}.

- a. 116
- b. 121
- c. 130
- d. 142

Question 2

Two independent events AA and BB are defined as P(A)=0.63, P(B)=0.47$P(A)$=0.63, $P(B)$=0.47, find P(A∩B)$P(A \cap B)$.

- a. 0.2959
- b. 0.2961
- c. 0.2960
- d. 0.2958

Question 3

Which measure of variation is the square root of the variance?

- a. The range
- b. The standard deviation
- c. The interquartile range
- d. The mean

Question 4

Two mutually exclusive events AA and BB are defined as P(A)=0.35$P(A)$=0.35, P(B)=0.25$P(B)$=0.25, find P(A|B)$P(A|B)$.

- a. 0.0875
- b. 0.0000
- c. 0.6000
- d. 0.1000

Question 5

Two independent events AA and BB are defined as P(A)=0.82, P(B)=0.37$P(A)$=0.82, $P(B)$=0.37, find P(A∩B)$P(A \cap B)$.

- a. 0.3024
- b. 0.3034
- c. 0.3044
- d. 0.3014

Question 6

The probability of two events AA and BB are defined by P(A)=0.425$P(A)$=0.425, P(B)=0.583$P(B)$=0.583, P(A∩B)=0.2478$P(A \cap B)$=0.2478, find P(A|B)$P(A|B)$.

- a. 0.4249
- b. 0.4252
- c. 0.4251
- d. 0.4254

Question 7

In a survey of 200 people, 75 said they preferred tea over coffee. What is the relative frequency of people who prefer tea?

- a. 0.7500
- b. 0.2667
- c. 0.3750
- d. 0.6250

Question 8

Two mutually exclusive events AA and BB are defined as P(A)=0.61$P(A)$=0.61, P(B)=0.28$P(B)$=0.28, find P(B|A)$P(B|A)$.

- a. 0.1708
- b. 0.4590
- c. 0.0000
- d. 0.2800

Question 9

For two events AA and BB, P(A)=0.594$P(A)$=0.594, P(B)=0.312$P(B)$=0.312 determine the value for which AA and BB are independent.

- a. 0.1854
- b. 0.1853
- c. 0.1855
- d. 0.1852

Question 10

Evaluate ∑(2x−1)$\sum(2x-1)$ for the sample data {2,4,6,8,10}$\{2,4,6,8,10\}$.

- a. 55
- b. 65
- c. 75
- d. 85

Question 11

Two independent events AA and BB are defined as P(A)=0.91, P(B)=0.44$P(A)=0.91$, $P(B)=0.44$, find P(A∩B)$P(A \cap B)$.

- a. 0.3994
- b. 0.4004
- c. 0.4014
- d. 0.3984

Question 12

What does the mean measure in a dataset?

- a. The spread of the scores
- b. The most frequent score
- c. The central tendency of the scores
- d. The difference between the highest and lowest scores

Question 13

Two mutually exclusive events AA and BB are defined as P(A)=0.52$P(A)=0.52$, P(B)=0.33$P(B)=0.33$, find P(A|B)$P(A|B)$.

- a. 0.1716
- b. 0.0000
- c. 0.6346
- d. 0.3300

Question 14

Two independent events AA and BB are defined as P(A)=0.28, P(B)=0.65$P(A)=0.28$, $P(B)=0.65$, find P(A∩B)$P(A \cap B)$.

- a. 0.1820
- b. 0.1830
- c. 0.1840
- d. 0.1810

Question 15

The probability of two events AA and BB are defined by P(A)=0.715$P(A)=0.715$, P(B)=0.487$P(B)=0.487$, P(A∩B)=0.3482$P(A \cap B)=0.3482$, find P(A|B)$P(A|B)$.

- a. 0.7148
- b. 0.7152
- c. 0.7150
- d. 0.7154

Question 16

In a class of 60 students, 18 scored an A grade. What is the relative frequency of students who scored an A?

- a. 0.3333
- b. 0.3000
- c. 0.2500
- d. 0.2000

Question 17

Two mutually exclusive events AA and BB are defined as P(A)=0.77$P(A)$=0.77, P(B)=0.14$P(B)$=0.14, find P(B|A)$P(B|A)$.

- a. 0.1078
- b. 0.0000
- c. 0.1818
- d. 0.1400

Question 18

For two events AA and BB, P(A)=0.806$P(A)$=0.806, P(B)=0.293$P(B)$=0.293 determine the value for which AA and BB are independent.

- a. 0.2362
- b. 0.2361
- c. 0.2363
- d. 0.2360

Question 19

Evaluate ∑(x2−3x)∑(x2−3x) for the sample data {1,3,5,7}{1,3,5,7}.

- a. 32
- b. 42
- c. 52
- d. 62

Question 20

Two independent events AA and BB are defined as P(A)=0.45, P(B)=0.84$P(A)$=0.45, $P(B)$=0.84, find P(A∩B)$P(A\cap B)$.

- a. 0.3770
- b. 0.3780
- c. 0.3790
- d. 0.3760

Question 21

If the variance of a dataset is 49, what is the standard deviation?

- a. 2401
- b. 7
- c. 49
- d. 24.5

Question 22

Two mutually exclusive events AA and BB are defined as P(A)=0.19$P(A)$=0.19, P(B)=0.46$P(B)$=0.46, find P(A|B)$P(A|B)$.

- a. 0.0874
- b. 0.0000
- c. 0.4130
- d. 0.4600

Question 23

Two independent events AA and BB are defined as P(A)=0.71, P(B)=0.58$P(A)$=0.71, $P(B)$=0.58, find P(A∩B)$P(A\cap B)$.

- a. 0.4118
- b. 0.4128
- c. 0.4138
- d. 0.4108

Question 24

The probability of two events AA and BB are defined by P(A)=0.592$P(A)$=0.592, P(B)=0.384$P(B)$=0.384, P(A∩B)=0.2273$P(A\cap B)$=0.2273, find P(A|B)$P(A|B)$.

- a. 0.5919
- b. 0.5921
- c. 0.5923
- d. 0.5917

Question 25

From a shipment of 500 items, 35 were found to be defective. What is the relative frequency of defective items?

- a. 0.0700
- b. 0.3500
- c. 0.0650
- d. 0.0750

Question 26

Two mutually exclusive events AA and BB are defined as P(A)=0.43$P(A)$=0.43, P(B)=0.51$P(B)$=0.51, find P(B|A)$P(B|A)$.

- a. 0.2193
- b. 0.0000
- c. 1.1860
- d. 0.5100

Question 27

For two events AA and BB, P(A)=0.285$P(A)$=0.285, P(B)=0.767$P(B)$=0.767 determine the value for which AA and BB are independent.

- a. 0.2186
- b. 0.2187
- c. 0.2185
- d. 0.2188

Question 28

Evaluate ∑(5−x2)∑(5−x2) for the sample data {2,4,6,8}{2,4,6,8}.

- a. -60
- b. -50
- c. -40
- d. -30

Question 29

Two independent events AA and BB are defined as P(A)=0.37, P(B)=0.92$P(A)$=0.37, $P(B)$=0.92, find P(A∩B)$P(A\cap B)$.

- a. 0.3394
- b. 0.3404
- c. 0.3414
- d. 0.3384

Question 30

The sum of all deviations from the mean in a dataset is always:

- a. Equal to the range
- b. Zero
- c. Positive
- d. Equal to the variance

Question 31

Two mutually exclusive events AA and BB are defined as P(A)=0.68$P(A)$=0.68, P(B)=0.27$P(B)$=0.27, find P(A|B)$P(A|B)$.

- a. 0.1836
- b. 0.0000
- c. 0.3971
- d. 0.2700

Question 32

Two independent events AA and BB are defined as P(A)=0.15, P(B)=0.73$P(A)$=0.15, $P(B)$=0.73, find P(A∩B)$P(A \cap B)$.

- a. 0.1085
- b. 0.1095
- c. 0.1105
- d. 0.1075

Question 33

The probability of two events AA and BB are defined by P(A)=0.963$P(A)$=0.963, P(B)=0.115$P(B)$=0.115, P(A∩B)=0.1107$P(A \cap B)$=0.1107, find P(A|B)$P(A|B)$.

- a. 0.9625
- b. 0.9627
- c. 0.9628
- d. 0.9626

Question 34

In a sample of 80 families, 32 have a pet dog. What is the relative frequency of families with a dog?

- a. 0.2500
- b. 0.4000
- c. 0.3200
- d. 0.5000

Question 35

Two mutually exclusive events AA and BB are defined as P(A)=0.24$P(A)$=0.24, P(B)=0.69$P(B)$=0.69, find P(B|A)$P(B|A)$.

- a. 0.1656
- b. 0.0000
- c. 0.3478
- d. 0.6900

Question 36

For two events AA and BB, P(A)=0.417$P(A)$=0.417, P(B)=0.692$P(B)$=0.692 determine the value for which AA and BB are independent.

- a. 0.2885
- b. 0.2886
- c. 0.2887
- d. 0.2884

Question 37

Evaluate ∑(3x+2)∑(3x+2) for the sample data {0,2,4,6,8}{0,2,4,6,8}.

- a. 50
- b. 60
- c. 70
- d. 80

Question 38

Two independent events AA and BB are defined as P(A)=0.68, P(B)=0.53$P(A)$=0.68, $P(B)$=0.53, find P(A∩B)$P(A\cap B)$.

- a. 0.3604
- b. 0.3614
- c. 0.3624
- d. 0.3594

Question 39

A data set has a mean of 50. If every score is increased by 5, what is the new mean?

- a. 50
- b. 55
- c. 250
- d. 45

Question 40

Two mutually exclusive events AA and BB are defined as P(A)=0.81$P(A)$=0.81, P(B)=0.19$P(B)$=0.19, find P(A|B)$P(A|B)$.

- a. 0.1539
- b. 0.0000
- c. 0.2346
- d. 0.1900

Question 41

Two independent events AA and BB are defined as P(A)=0.42, P(B)=0.65$P(A)$=0.42, $P(B)$=0.65, find P(A∩B)$P(A\cap B)$.

- a. 0.2730
- b. 0.2740
- c. 0.2750
- d. 0.2720

Question 42

The probability of two events AA and BB are defined by P(A)=0.214$P(A)$=0.214, P(B)=0.825$P(B)$=0.825, P(A∩B)=0.1766$P(A \cap B)$=0.1766, find P(A|B)$P(A|B)$.

- a. 0.2140
- b. 0.2141
- c. 0.2142
- d. 0.2143

Question 43

In a bag of 120 candies, 48 are red. What is the relative frequency of red candies?

- a. 0.4800
- b. 0.4000
- c. 0.2500
- d. 0.2000

Question 44

Two mutually exclusive events AA and BB are defined as P(A)=0.33$P(A)$=0.33, P(B)=0.57$P(B)$=0.57, find P(B|A)$P(B|A)$.

- a. 0.1881
- b. 0.0000
- c. 1.7273
- d. 0.5700

Question 45

For two events AA and BB, P(A)=0.672$P(A)$=0.672, P(B)=0.459$P(B)$=0.459 determine the value for which AA and BB are independent.

- a. 0.3085
- b. 0.3084
- c. 0.3086
- d. 0.3083

Question 46

Evaluate ∑(4x−x2)∑(4x−x2) for the sample data {1,2,3,4}{1,2,3,4}.

- a. 10

- b. 0
- c. -10
- d. 20

Question 47

Two independent events AA and BB are defined as P(A)=0.59, P(B)=0.86$P(A)=0.59$, $P(B)=0.86$, find P(A∩B)$P(A\cap B)$.

- a. 0.5064
- b. 0.5074
- c. 0.5084
- d. 0.5054

Question 48

If the range of a dataset is 25 and the smallest value is 10, what is the largest value?

- a. 15
- b. 35
- c. 2.5
- d. 250

Question 49

Two mutually exclusive events AA and BB are defined as P(A)=0.44$P(A)=0.44$, P(B)=0.39$P(B)=0.39$, find P(A|B)$P(A|B)$.

- a. 0.1716
- b. 0.0000
- c. 1.1282
- d. 0.3900

Question 50

Two independent events AA and BB are defined as P(A)=0.11, P(B)=0.88$P(A)=0.11$, $P(B)=0.88$, find P(A∩B)$P(A\cap B)$.

- a. 0.0958
- b. 0.0968
- c. 0.0978
- d. 0.0948

Question 51

The probability of two events AA and BB are defined by P(A)=0.352$P(A)=0.352$, P(B)=0.946$P(B)=0.946$, P(A∩B)=0.3328$P(A\cap B)=0.3328$, find P(A|B)$P(A|B)$.

- a. 0.3518

- b. 0.3519
- c. 0.3520
- d. 0.3521

Question 52

In a group of 25 people, 10 are left-handed. What is the relative frequency of right-handed people?

- a. 0.4000
- b. 0.6000
- c. 0.1000
- d. 2.5000

Question 53

Two mutually exclusive events AA and BB are defined as P(A)=0.08$P(A)$=0.08, P(B)=0.75$P(B)$=0.75, find P(B|A)$P(B|A)$.

- a. 0.0600
- b. 0.0000
- c. 0.1067
- d. 0.7500

Question 54

For two events AA and BB, P(A)=0.912$P(A)$=0.912, P(B)=0.237$P(B)$=0.237 determine the value for which AA and BB are independent.

- a. 0.2161
- b. 0.2162
- c. 0.2163
- d. 0.2160

Question 55

Evaluate ∑(x2+4x)∑(x2+4x) for the sample data {−1,0,1,2}{−1,0,1,2}.

- a. 14
- b. 18
- c. 22
- d. 26

Question 56

Two independent events AA and BB are defined as P(A)=0.33, P(B)=0.77$P(A)$=0.33, $P(B)$=0.77, find P(A∩B)$P(A\cap B)$.

- a. 0.2541
- b. 0.2551

- c. 0.2561
- d. 0.2531

Question 57

Which measure of central tendency is most affected by extreme scores (outliers)?

- a. The median
- b. The mode
- c. The mean
- d. The range

Question 58

Two mutually exclusive events AA and BB are defined as P(A)=0.62$P(A)$=0.62, P(B)=0.34$P(B)$=0.34, find P(A|B)$P(A|B)$.

- a. 0.2108
- b. 0.0000
- c. 0.5484
- d. 0.3400

Question 59

Two independent events AA and BB are defined as P(A)=0.55, P(B)=0.49$P(A)$=0.55, $P(B)$=0.49, find P(A∩B)$P(A\cap B)$.

- a. 0.2695
- b. 0.2705
- c. 0.2715
- d. 0.2685

Question 60

The probability of two events AA and BB are defined by P(A)=0.871$P(A)$=0.871, P(B)=0.163$P(B)$=0.163, P(A∩B)=0.1420$P(A\cap B)$=0.1420, find P(A|B)$P(A|B)$.

- a. 0.8712
- b. 0.8711
- c. 0.8710
- d. 0.8713

Question 61

Out of 1500 seeds planted, 1200 germinated. What is the relative frequency of germination?

- a. 0.2000
- b. 0.8000

- c. 1.2500
- d. 0.1200

Question 62

Two mutually exclusive events AA and BB are defined as P(A)=0.13$P(A)$=0.13, P(B)=0.48$P(B)$=0.48, find P(B|A)$P(B|A)$.

- a. 0.0624
- b. 0.0000
- c. 0.2708
- d. 0.4800

Question 63

For two events AA and BB, P(A)=0.184$P(A)$=0.184, P(B)=0.735$P(B)$=0.735 determine the value for which AA and BB are independent.

- a. 0.1352
- b. 0.1353
- c. 0.1354
- d. 0.1351

Question 64

Evaluate ∑(2x2−3)∑(2x2−3) for the sample data {2,3,4}{2,3,4}.

- a. 52
- b. 55
- c. 58
- d. 61

Question 65

Two independent events AA and BB are defined as P(A)=0.86, P(B)=0.31$P(A)$=0.86, $P(B)$=0.31, find P(A∩B)$P(A\cap B)$.

- a. 0.2656
- b. 0.2666
- c. 0.2676
- d. 0.2646

Question 66

For a symmetric distribution, the mean and median are:

- a. Always zero
- b. Equal
- c. Different
- d. Equal to the mode only

Question 67

Two mutually exclusive events AA and BB are defined as P(A)=0.97$P(A)$=0.97, P(B)=0.02$P(B)$=0.02, find P(A|B)$P(A|B)$.

- a. 0.0194
- b. 0.0000
- c. 0.0206
- d. 0.0200

Question 68

Two independent events AA and BB are defined as P(A)=0.22, P(B)=0.69$P(A)$=0.22, $P(B)$=0.69, find P(A∩B)$P(A\cap B)$.

- a. 0.1518
- b. 0.1528
- c. 0.1538
- d. 0.1508

Question 69

The probability of two events AA and BB are defined by P(A)=0.627$P(A)$=0.627, P(B)=0.208$P(B)$=0.208, P(A∩B)=0.1303$P(A\cap B)$=0.1303, find P(A|B)$P(A|B)$.

- a. 0.6266
- b. 0.6267
- c. 0.6268
- d. 0.6269

Question 70

A die is rolled 90 times and the number 3 appears 12 times. What is the relative frequency of rolling a 3?

- a. 0.1333
- b. 0.1200
- c. 0.3333
- d. 7.5000

Question 71

Two mutually exclusive events AA and BB are defined as P(A)=0.39$P(A)$=0.39, P(B)=0.41$P(B)$=0.41, find P(B|A)$P(B|A)$.

- a. 0.1599
- b. 0.0000
- c. 1.0513

- d. 0.4100

Question 72

For two events AA and BB, P(A)=0.502$P(A)$=0.502, P(B)=0.682$P(B)$=0.682 determine the value for which AA and BB are independent.

- a. 0.3423
- b. 0.3424
- c. 0.3425
- d. 0.3422

Question 73

Evaluate ∑(5x−x2)∑(5x−x2) for the sample data {1,3,5}{1,3,5}.

- a. 14
- b. 15
- c. 16
- d. 17

Question 74

Two independent events AA and BB are defined as P(A)=0.41, P(B)=0.82$P(A)$=0.41, $P(B)$=0.82, find P(A∩B)$P(A∩B)$.

- a. 0.3362
- b. 0.3372
- c. 0.3382
- d. 0.3352

Question 75

What is the primary difference between variance and standard deviation?

- a. Variance is the square of the standard deviation.
- b. Standard deviation is the square of the variance.
- c. They measure different things.
- d. Variance is always larger.

Question 76

Two mutually exclusive events AA and BB are defined as P(A)=0.55$P(A)$=0.55, P(B)=0.44$P(B)$=0.44, find P(A|B)$P(A|B)$.

- a. 0.2420
- b. 0.0000
- c. 0.8000
- d. 0.4400

Question 77

Two independent events AA and BB are defined as P(A)=0.19, P(B)=0.67$P(A)$=0.19, $P(B)$=0.67, find P(A∩B)$P(A \cap B)$.

- a. 0.1273
- b. 0.1283
- c. 0.1293
- d. 0.1263

Question 78

The probability of two events AA and BB are defined by P(A)=0.485$P(A)$=0.485, P(B)=0.256$P(B)$=0.256, P(A∩B)=0.1242$P(A \cap B)$=0.1242, find P(A|B)$P(A|B)$.

- a. 0.4852
- b. 0.4851
- c. 0.4850
- d. 0.4853

Question 79

In a study, 36 out of 120 participants were male. What is the relative frequency of female participants?

- a. 0.3000
- b. 0.7000
- c. 0.3600
- d. 0.6400

Question 80

Two mutually exclusive events AA and BB are defined as P(A)=0.18$P(A)$=0.18, P(B)=0.52$P(B)$=0.52, find P(B|A)$P(B|A)$.

- a. 0.0936
- b. 0.0000
- c. 0.3462
- d. 0.5200

Question 81

For two events AA and BB, P(A)=0.369$P(A)$=0.369, P(B)=0.841$P(B)$=0.841 determine the value for which AA and BB are independent.

- a. 0.3103
- b. 0.3104
- c. 0.3105
- d. 0.3102

Question 82

Evaluate ∑(x2−4x+4)∑(x2−4x+4) for the sample data {0,2,4}{0,2,4}.

- a. 8
- b. 12
- c. 16
- d. 20

Question 83

Two independent events AA and BB are defined as P(A)=0.73, P(B)=0.46$P(A)$=0.73, $P(B)$=0.46, find P(A∩B)$P(A∩B)$.

- a. 0.3358
- b. 0.3368
- c. 0.3378
- d. 0.3348

Question 84

A dataset has a mean of 100. If every score is multiplied by 2, what is the new mean?

- a. 50
- b. 100
- c. 200
- d. 102

Question 85

Two mutually exclusive events AA and BB are defined as P(A)=0.42$P(A)$=0.42, P(B)=0.36$P(B)$=0.36, find P(A|B)$P(A|B)$.

- a. 0.1512
- b. 0.0000
- c. 0.8571
- d. 0.3600

Question 86

Two independent events AA and BB are defined as P(A)=0.64, P(B)=0.55$P(A)$=0.64, $P(B)$=0.55, find P(A∩B)$P(A∩B)$.

- a. 0.3520
- b. 0.3530
- c. 0.3540
- d. 0.3510

Question 87

The probability of two events AA and BB are defined by P(A)=0.123$P(A)$=0.123, P(B)=0.456$P(B)$=0.456, P(A∩B)=0.0561$P(A \cap B)$=0.0561, find P(A|B)$P(A|B)$.

- a. 0.1230
- b. 0.1231
- c. 0.1232
- d. 0.1229

Question 88

A coin is flipped 200 times and lands on heads 108 times. What is the relative frequency of tails?

- a. 0.5400
- b. 0.4600
- c. 0.5000
- d. 0.5400

Question 89

Two mutually exclusive events AA and BB are defined as P(A)=0.66$P(A)$=0.66, P(B)=0.27$P(B)$=0.27, find P(B|A)$P(B|A)$.

- a. 0.1782
- b. 0.0000
- c. 0.4091
- d. 0.2700

Question 90

For two events AA and BB, P(A)=0.628$P(A)$=0.628, P(B)=0.515$P(B)$=0.515 determine the value for which AA and BB are independent.

- a. 0.3234
- b. 0.3235
- c. 0.3236
- d. 0.3233

Question 91

Evaluate ∑(x3−x)∑(x3−x) for the sample data {1,2,3}{1,2,3}.

- a. 30
- b. 32
- c. 34
- d. 36

Question 92

Two independent events AA and BB are defined as P(A)=0.47, P(B)=0.79$P(A)=0.47$, $P(B)=0.79$, find P(A∩B)$P(A \cap B)$.

- a. 0.3703
- b. 0.3713
- c. 0.3723
- d. 0.3693

Question 93

If the mean of a dataset is 25 and the standard deviation is 5, what is the variance?

- a. 5
- b. 25
- c. 125
- d. 30

Question 94

Two mutually exclusive events AA and BB are defined as P(A)=0.87$P(A)=0.87$, P(B)=0.05$P(B)=0.05$, find P(A|B)$P(A|B)$.

- a. 0.0435
- b. 0.0000
- c. 0.0575
- d. 0.0500

Question 95

Two independent events AA and BB are defined as P(A)=0.26, P(B)=0.81$P(A)=0.26$, $P(B)=0.81$, find P(A∩B)$P(A \cap B)$.

- a. 0.2096
- b. 0.2106
- c. 0.2116
- d. 0.2086

Question 96

The probability of two events AA and BB are defined by P(A)=0.753$P(A)=0.753$, P(B)=0.234$P(B)=0.234$, P(A∩B)=0.1761$P(A \cap B)=0.1761$, find P(A|B)$P(A|B)$.

- a. 0.7526
- b. 0.7527
- c. 0.7528
- d. 0.7525

Question 97

In a production line, 15 out of 500 items were flawed. What is the relative frequency of non-flawed items?

- a. 0.0300
- b. 0.9700
- c. 0.1500
- d. 0.8500

Question 98

Two mutually exclusive events AA and BB are defined as P(A)=0.31$P(A)$=0.31, P(B)=0.47$P(B)$=0.47, find P(B|A)$P(B|A)$.

- a. 0.1457
- b. 0.0000
- c. 1.5161
- d. 0.4700

Question 99

For two events AA and BB, P(A)=0.194$P(A)$=0.194, P(B)=0.532$P(B)$=0.532 determine the value for which AA and BB are independent.

- a. 0.1032
- b. 0.1033
- c. 0.1034
- d. 0.1031

Question 100

Evaluate ∑(3x2−2x)∑(3x2−2x) for the sample data {0,1,2,3}{0,1,2,3}.

- a. 32
- b. 34
- c. 36
- d. 38